Module 55

Backup & Recovery/5: Backup/2: RAID

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Module Recap

- Recovery based on operation logging supplements log-based recovery

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- Planning for Backup

Module Objectives

- Understanding RAID: Array of redundant disks in parallel to enhance speed and reliability

Module Outline

- Redundant Array of Independent Disks: RAID

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  - Except for dependent failure modes such as fire or building collapse or electrical power surges

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Limitations

While most RAID levels can provide good protection against and recovery from hardware defects or defective sectors/read errors (hard errors), they do not provide any protection against data loss due to catastrophic failures (fire, water) or soft errors such as user error, software malfunction, or malware infection. For valuable data, RAID is only one building block

RAID 0: Striping

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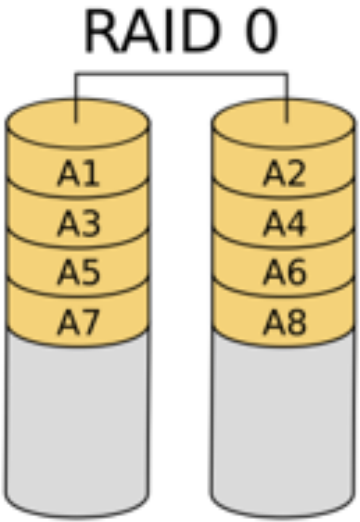

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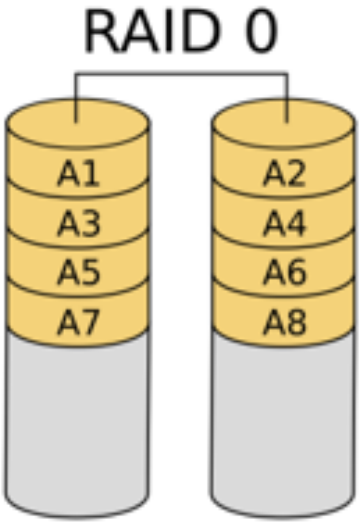

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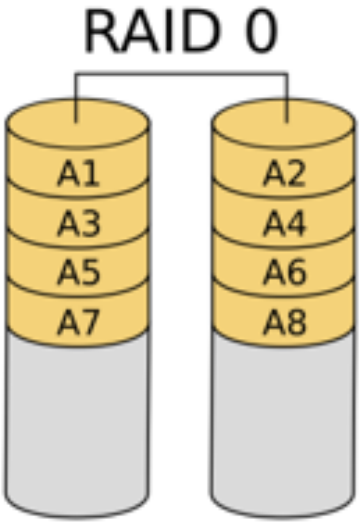

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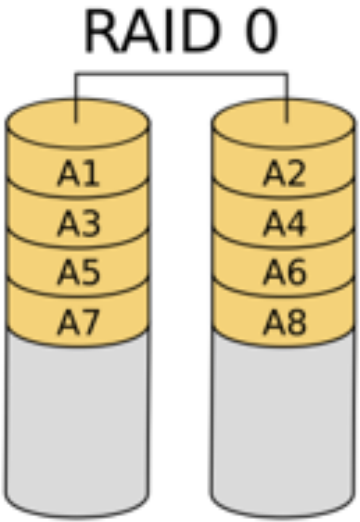

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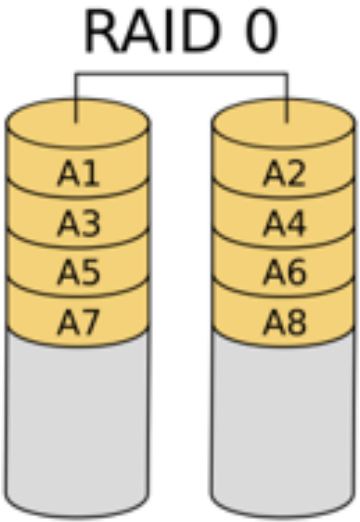

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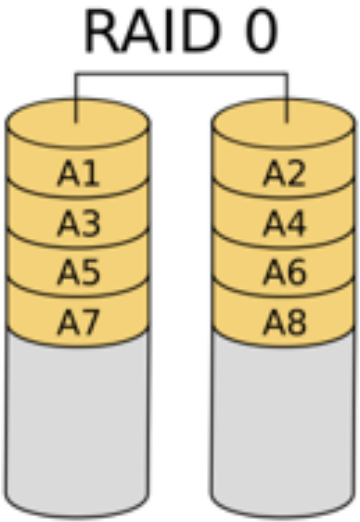

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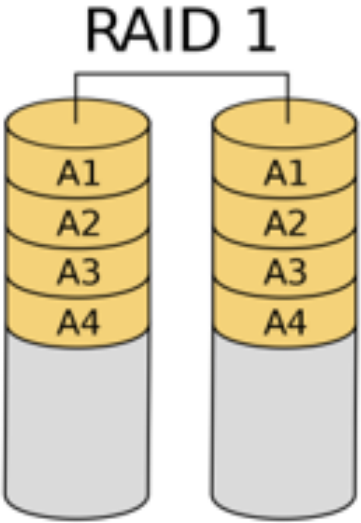

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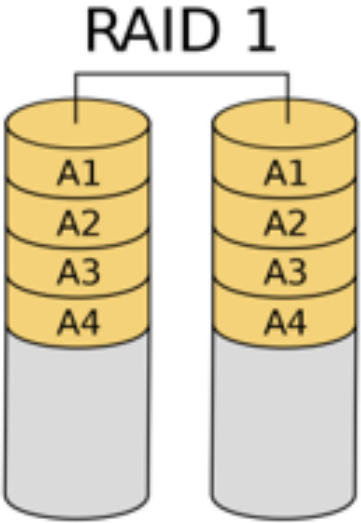

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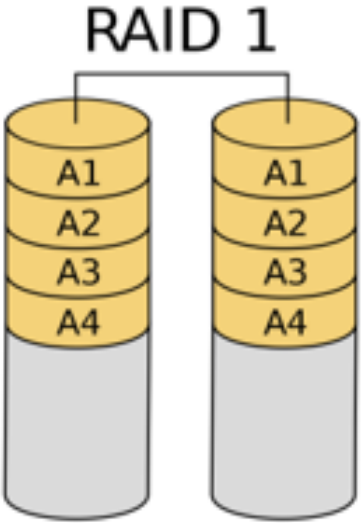

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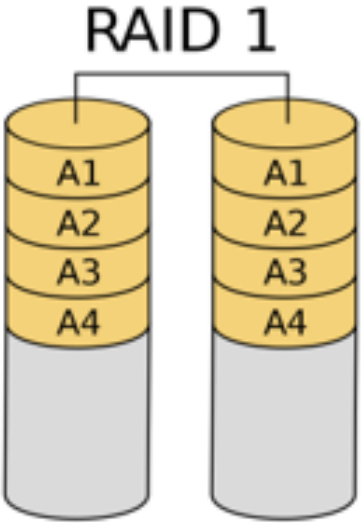

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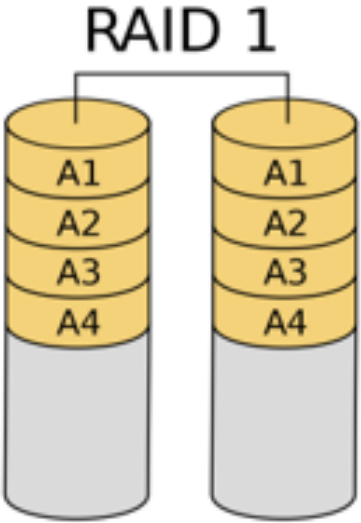

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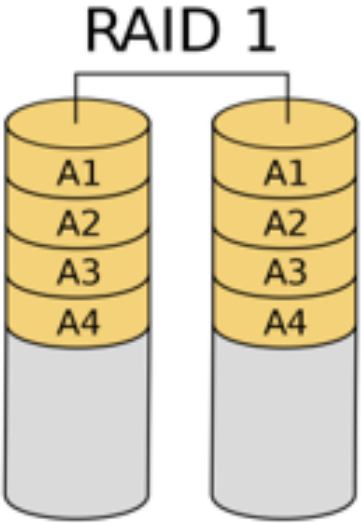

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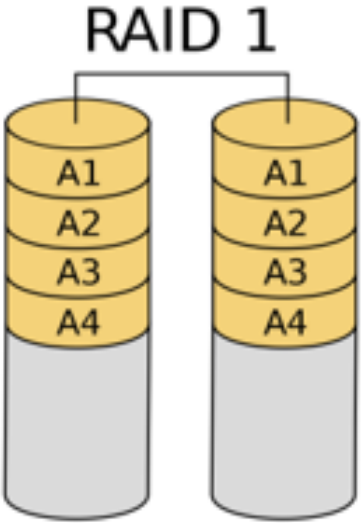

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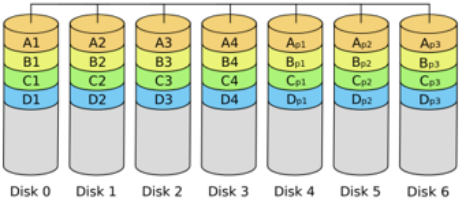

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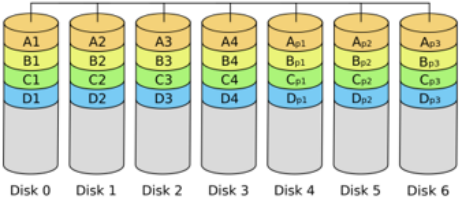

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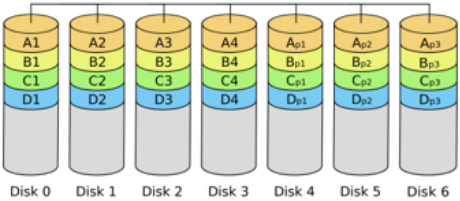

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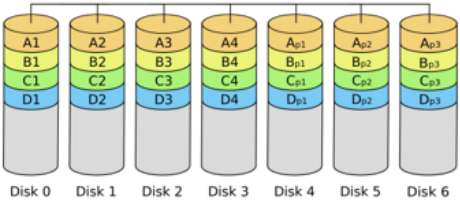

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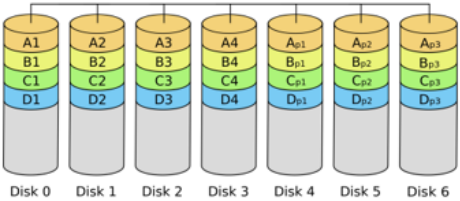

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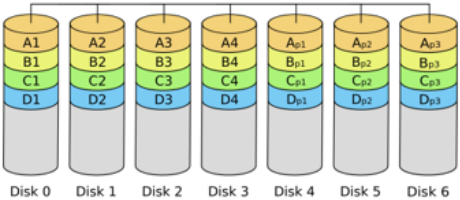

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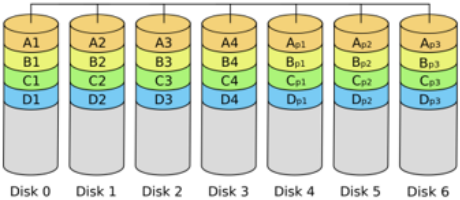

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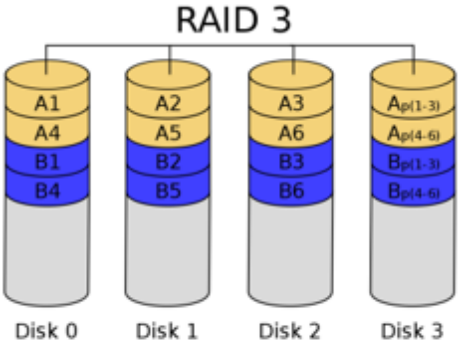

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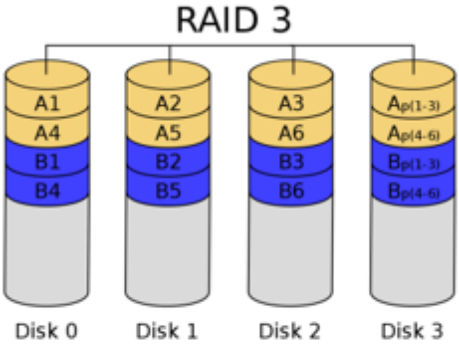

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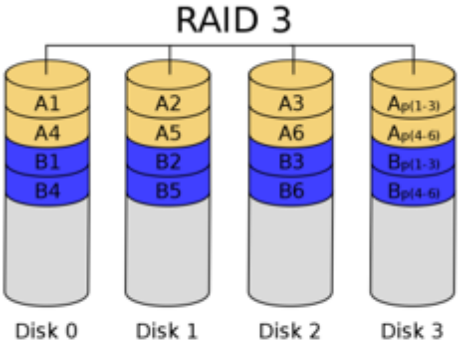

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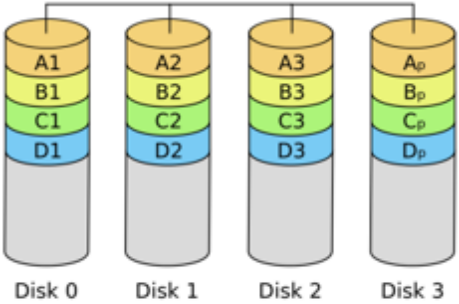

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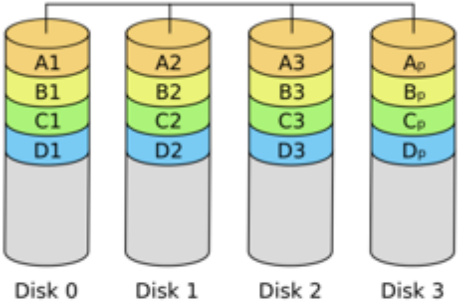

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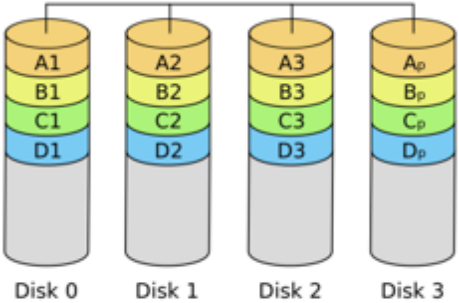

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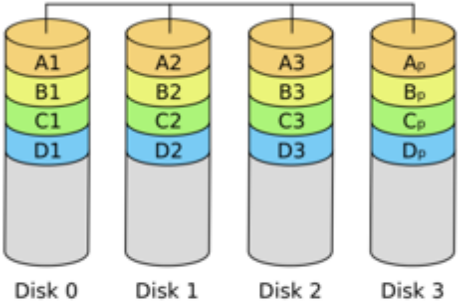

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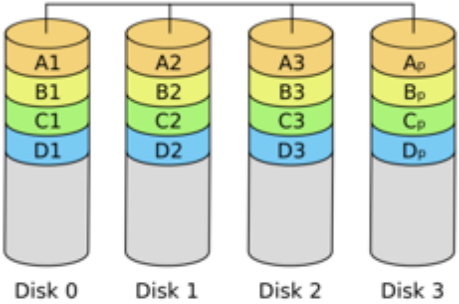

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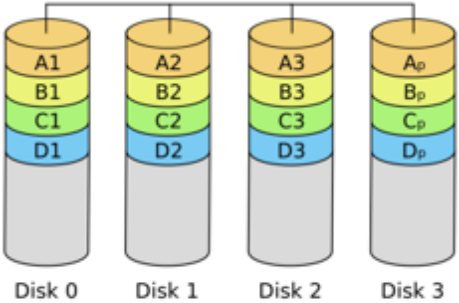

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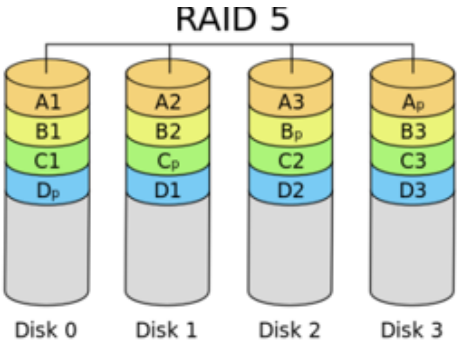

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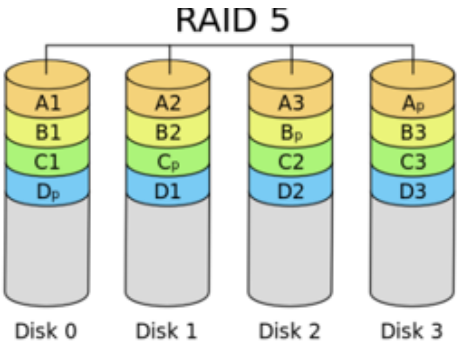

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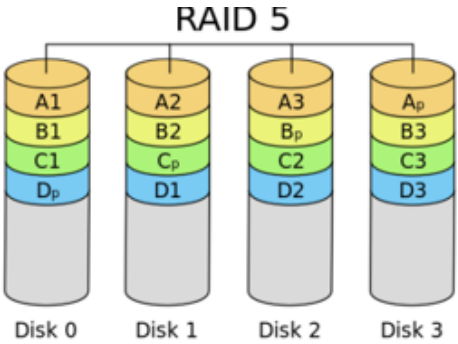

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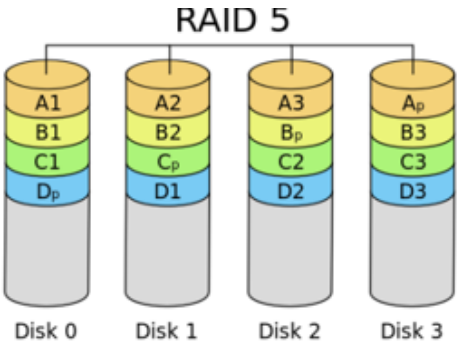

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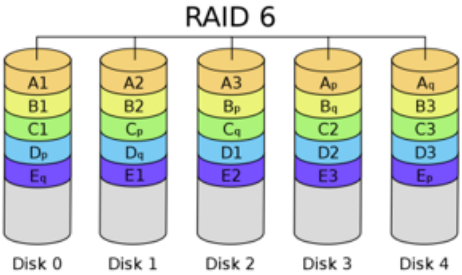

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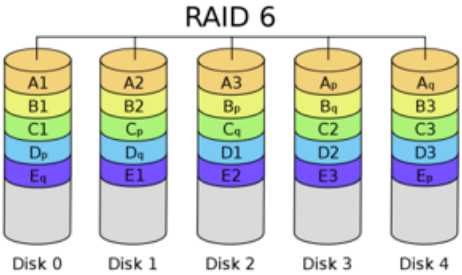

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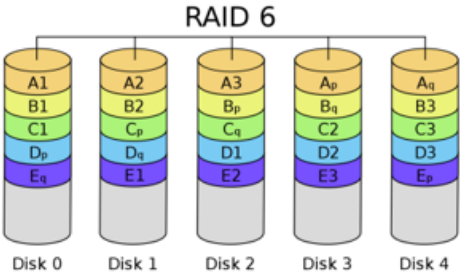

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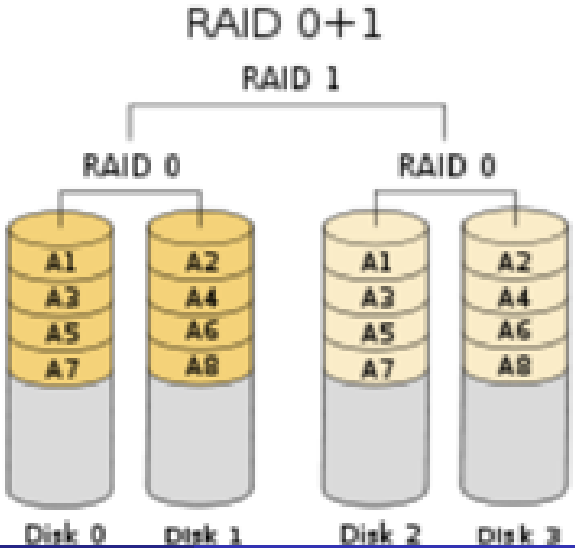
- As a result of the layering scheme, RAID 01 and RAID 10 represent significantly different nested RAID levels

RAID 01 (RAID 0+1): Mirror of Stripes

- RAID 01 is a **mirror of stripes**



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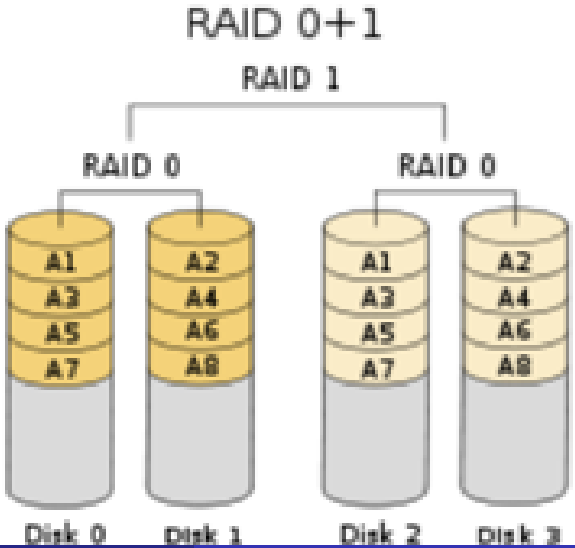

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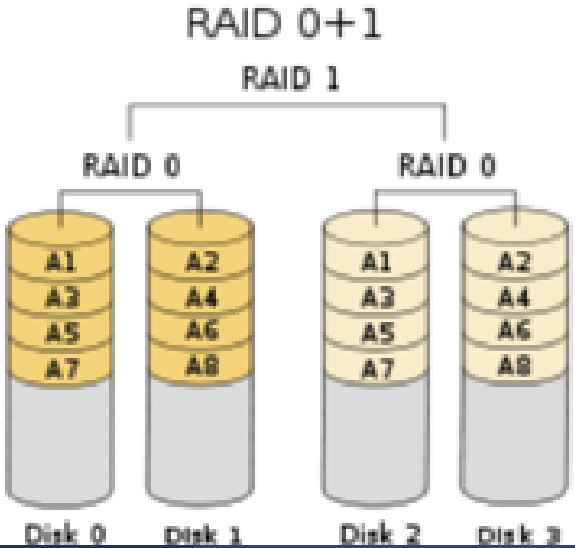

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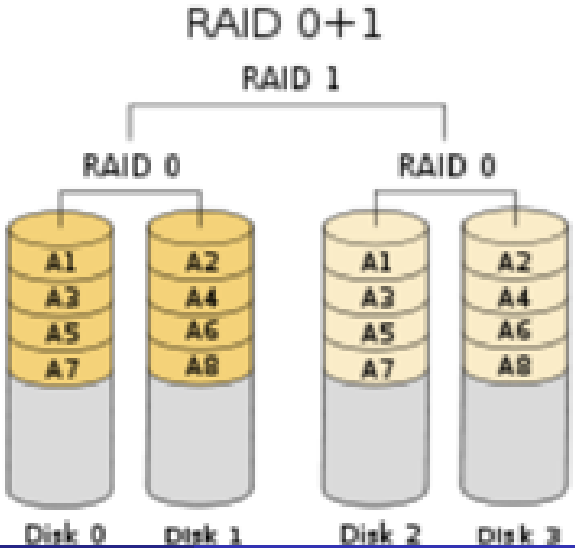

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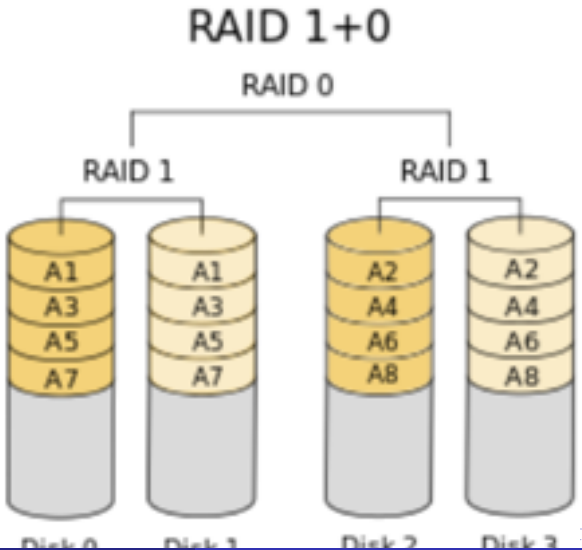

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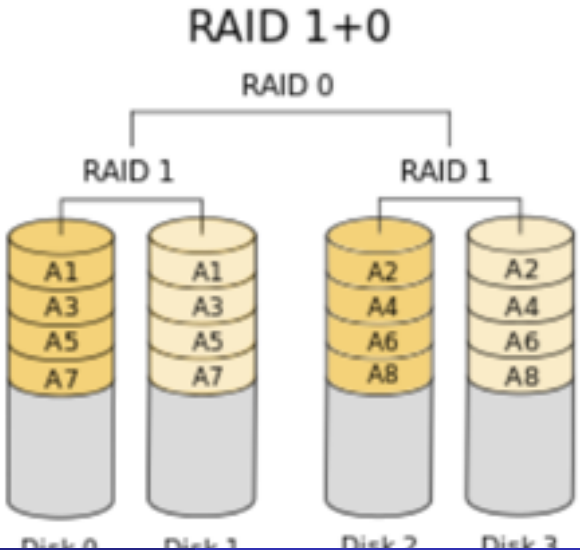

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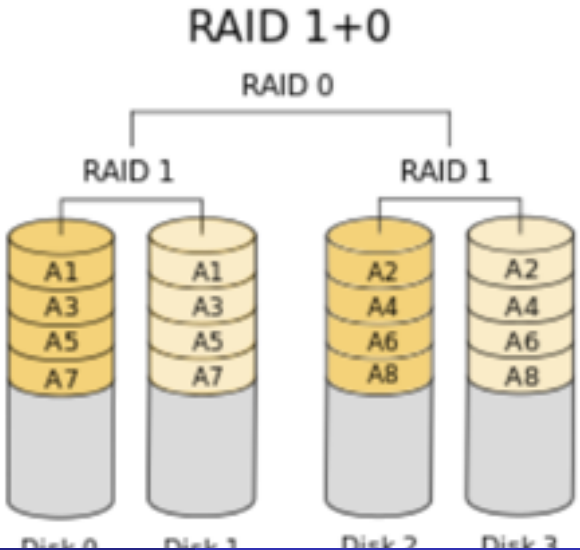

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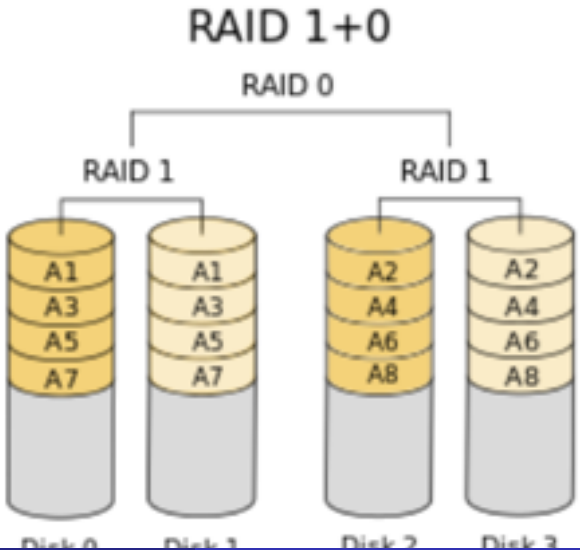
- Thus, it is the preferable RAID level for I/O-intensive applications such as database, email, and web servers, as well as for any other use requiring high disk performance



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Comparison of RAID: Theoretical

Level	Description	Min. drives	Space Eff.	Fault Tolerance (Drives)	Perf: Read	Perf: Write
RAID 0	Block striping w/o parity or mirror	2	1	None	n	n
RAID 1	Mirroring w/o parity or striping	2	$1/n$	$n-1$	n	1
RAID 2	Bit striping with Hamming code	3	$1-1/n\lg(n+1)$	One	Depends	Depends
RAID 3	Byte striping with dedicated parity	3	$1-1/n$	One	$n-1$	$n-1$
RAID 4	Block striping with dedicated parity	3	$1-1/n$	One	$n-1$	$n-1$
RAID 5	Block striping with dist. parity	3	$1-1/n$	One	n	single: $1/4$, stripe: $n-1$
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Note

n: factor of single disk. Performance assumes hardware capable of performing calculations fast enough.

Comparison of RAID: Practical

Features	**RAID 0**	**RAID 1**	**RAID 5**	**RAID 6**	**RAID 10**
Min. drives	2	2	3	4	4
Fault tolerance	None	Single-drive	Single-drive	Two-drive	Up to 1 disk in each sub-array
Read perf.	High	Medium	Low	Low	High
Write perf.	High	Medium	Low	Low	Medium
Capacity util.	100%	50%	67% - 94%	50% - 88%	50%
Typical apps.	High end workstations	Operating systems	Data warehouse	Data archive	Fast databases

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